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Aqal: First Urdu Reasoning Large Language Model through Continued Pretraining and GRPO-Based Reinforcement Learning

Anonymous ACL submission

1 Evaluation Methodology

1.1 Metrics and Why Each Is Necessary

One score on accuracy is not sufficient to a reasoning model- it lumps failures of very different causes into a single score. Four measures help us to provide a complete picture.

Exact Match (EM). EM tests the model whether the answer of the model matches the reference. It is our major cue as it is objective and not dependent on human beings. Its strength is also a weakness: EM is unaware of the way the answer was arrived at, and thus when we compare it with step-level scores, we can tell that correctness is achieved either as a result of real thought or it was achieved by chance as the correct pattern is being filled in.

Step-Level Consistency. Two Urdu-speaking independent annotators verify that every step between the steps is logically correct before the next and is linguistically correct. This captures two types of failures used by EM that run off the calculator at the last arithmetic step suffers zero marks on EM even with perfectly sound logic; and vice versa, a model which makes up for plausible-looking steps that accidentally cancel to the correct answer scores full marks on EM even with a logic that is incoherent. These cases are divided by step consistency.

Perplexity. Throughout CPT, we follow perplexity on a held-out Urdu Wikipedia split. Unless CPT decreases the level of perplexity, the model has never learned to internalise the distribution of Urdu tokens, which implies that SFT and GRPO are constructed on a weak linguistic basis. Perplexity is a precondition test, and not a proxy of reasoning capacity.

Human Evaluation. Annotators rate reasoning clarity, logical soundness and frequency of hallucinations on 100 sampled outputs per condition. Automated measures are unable to capture minor language breakdown - when the model generates Urdu words but takes English syntax argument structure - or to even determine whether mathematical Urdu is natural as a native speaker would read it.

1.2 Ablation Design

Our four conditions form a factorial ablation where each comparison isolates exactly one variable. **B1** (Alif-1.0-8B-Instruct, zero-shot) is the lower bound. Alif scores 75.5 on general Urdu benchmarks (Alif Team, 2025), so its poor math performance reflects a reasoning alignment gap, not a language gap. **B2** (Alif, 8-shot chain-of-thought (Wei et al., 2022)) adds in-context learning with no parameter change, measuring the ceiling of pure prompting. **B3** (Qwen2.5-1.5B, CPT+SFT) changes both model and training. The B2→B3 comparison is deliberately cross-scale—1.5B vs. 8B—so any performance gap must come from the training stages, not from scale. **B4** (CPT+SFT+GRPO) adds reinforcement learning to B3 with everything else held fixed; the B3→B4 gap isolates GRPO exactly.

All EM results are averaged over three runs with seed 42 and tested with paired bootstrap (n=10,000, p<0.05). Variance at 1.5B scale on a T4 is non-trivial, so we report it alongside means rather than treating point estimates as definitive.

Condition	Params	EM (%)	Step (%)
B1: Alif zero-shot	8B	18.6	21.3
B2: Alif CoT	8B	26.4	38.7
B3: CPT+SFT	1.5B	31.2	44.1
B4: CPT+SFT+GRPO	1.5B	TBD	TBD
Target	—	≥45.0	≥60.0

Table 1: MATH-500-Urdu results. B3 outperforms the 5× larger B2, isolating instruction alignment over scale. B4 pending GRPO completion.

2 Error and Failure Taxonomy

We analysed 50 failure cases per condition and identified five failure types, each mapped to a specific training-stage deficit.

Type I — Correct Reasoning, Wrong Answer. The reasoning is valid but the ultimate conclusion is in error, usually an arithmetic error in the last stage. The consistency of the steps is high; EM = 0. The reason is a gradient imbalance in SFT: cross-entropy loss is distributed across the entire response, thus the final answer, which is much less tokens than the reasoning chain, gets a much weaker training signal. The first stage to punish this failure directly is the scalar correctness reward of GRPO.

Type II — Correct Answer, Incoherent Reasoning. The solution is correct yet the steps in between are internally contradictory or falsified. This is rated by EM as a perfect solution. The reason is that SFT artefact: the model is taught the format of chain-of-thought reasoning: numbered steps, calculation blocks, a conclusion because this format is present in training responses uniformly. The cross-entropy objective however does not impose any constraints that the steps themselves must be logically necessary. The model becomes trained to imitate the form of reasoning, and not its contents.

Type III — Hallucinated Steps. The model imposes quantities or premises that are not on the problem statement. This dates to CPT: language modelling is an incentive to continue a mathematical prose without checking it against any grounding. SFT decreases it in the case of problems that have been trained, but it still remains in the unfamiliar structure. GRPO indirectly suppresses it, by causing steps that have been fabricated to give it a false answer; it does not explicitly punish it.

Type IV — Language Collapse. The model is changed to English when there is a spike in mathematical complexity. Qwen2.5-1.5B was trained more on English and Chinese wherein English was the default register of hard mathematics. CPT changes the endogenous distribution of tokens to Urdu; SFT strengthens Urdu by monitoring it; GRPO involves the primary stage of enacting punishment (via a language consistency reinforcement element) to code-switching. The passive distribution shift in B1→B3 drop and active alignment in B3→B4 drop are examples of language collapse.

Type V — Shallow Pattern Matching. The model is right in recognising the category of surface operation, proportion, rate, but it uses a memorised template which fractures when there is a slight structural difference. This is a failure in diagnosis of GRPO: in case GRPO teaches transferable reasoning, Type V errors must occur in the more challenging subcategories of MATH-500. In case they continue doing so, GRPO is adapting the GSM8K distribution but not learning to reason.

3 Diagnostic Insights

3.1 SFT Improves Reasoning Without Math Data

The most unexpected result is the B2→B3 jump: a 1.5B model fine-tuned on a general instruction dataset outperforms an 8B model with chain-of-thought prompting on a mathematics benchmark, without seeing a single math training example. EM rises from 26.4% to 31.2% and step consistency from 38.7% to 44.1%.

The reason is that multi-step reasoning is a structural skill. Alpaca-GPT4-Urdu does not offer us knowledge of a mathematical nature, but a pattern of generation: specify a goal, generate steps that decrease it, and end at an endpoint. GPT-4 responses generated are internally consistent, explicitly tied with logical connectives and connect the premises to the conclusions. This mode of generation is fine-tuned to teach us a lesson applicable in mathematics since the structure underlying it, which is conditional sequential inference toward a goal, is generically the same. The low resource NLP implication is considerable: the domain specific annotation might not be expensive to achieve early reasoning benefits; general instruction high quality data might be enough to issue the structure. (Ouyang et al., 2022).

3.2 CPT Grounds Language But Cannot Teach Reasoning

CPT will decrease perplexity and inhibit language collapse but does not enhance reasoning in and of itself. The language modelling is a goal-free task, that is, you are supposed to predict the next token based on what you know of the context, and you are not expected to have a goal state and to check logical entailment. Wikipedia is filled with mathematical material, which is declarative prose, i.e. statements of reasoning and not a sequence of reasoning. CPT shows the model how to explain how

a solution would appear; SFT only teaches how to create one. However, CPT is an indispensable preparation: in its absence, SFT would be attempting to impart organized Urdu generation on a model whose own language structure and vocabulary has not internalised Urdu syntax and vocabulary, resulting in the language breakdown in B1.

3.3 GRPO: Genuine Reasoning or Reward Fitting?

GRPO replaces the SFT objective with reward maximisation over sampled completions (Shao et al., 2024). In contrast to loss based on cross-entropy, which broadens the signal signal across tokens, the reward flow back a single answer along the entire chain, causing the intermediate steps to causally justify the conclusion instead of being form-similar to correct steps. It is through this that GRPO would be able to repair Type I and Type II failures that are inaccessible by SFT.

Whether it does depends on training distribution diversity. GSM8K contains 8,500 elementary arithmetic problems with homogeneous solution structures (Cobbe et al., 2021). A narrow reward landscape risks converging to a policy that maximises reward on GSM8K-type problems without exploring the reasoning strategies that MATH-500 demands. We predict B4 will gain most on MATH-500's algebra and arithmetic subcategories—closest to GSM8K—and least on geometry, number theory, and calculus. A flat improvement profile would suggest genuine transfer; a skewed one would confirm distribution-local fitting.

3.4 The GSM8K-to-MATH-500 Transfer Gap

The mismatch between GRPO training (elementary arithmetic) and evaluation (competition mathematics (Hendrycks et al., 2021)) is intentional—it probes whether GRPO's gains are transferable or local. Any meaningful B3→B4 improvement on MATH-500 provides evidence for domain-general reasoning. A flat gap is also scientifically informative: it would show that narrow GRPO training data is the bottleneck, which directly motivates collecting harder Urdu RL training data for future work.

4 Limitations of Evaluation

EM Is Blind to Process. This is deliberate, though it has a more profound side effect: even the human evaluation at the level of step is not very precise. Local coherence is checked by annotators, which is, does this step build on the previous one, but this is not formal proof verification. A model might generate a plausible step that on its own is individually plausible, but when these steps are combined together they represent an invalid inference, which an annotator would most probably fail to detect. With no formal semantic grammar of Urdu mathematics, step scores are invariably inaccurate.

Translated Benchmark Artefacts. MATH-500-Urdu is translated into machines. The Urdu mathematical writing combines Arabic-Indic numerals, Latin notation, and English transliterated words in a manner depending on the tradition of the Indian subcontinent. The translation pipeline contains certain decisions that are not likely to reflect the writing of a student or an educator. A localized version scoring highly on the translation benchmark may keep facing difficulty with organically generated Urdu mathematics (Ahuja et al., 2023) and this restricts the ecological validity of our inferences.

Reward Proxy Misalignment. GRPO reward consists of three parts (answer correctness (exact string match), step completeness (structural Markers) and language consistency (Urdu character ratio). Each is a surface proxy. Exact string match will penalise an accurate answer in a different form to the reference. Structural markers are quantitative indicators of steps that are numbered, rather than those which are logically sound. Script rather than syntactic naturalness is determined by character ratio. A model is able to score high on all three and still give Type II outputs. This is one of the known weaknesses of reward-based alignment, which is further aggravated by the lack of a semantic checker of Urdu.

LoRA Rank Constraint. Running on a single T4 forces LoRA at rank 64. For CPT and SFT this is a reasonable trade-off. For GRPO it is more restrictive: policy gradient updates during exploration require representing a wider distribution of completions, and rank 64 may cap the adapter's expressiveness. B4 results should be treated as a lower bound on what GRPO could achieve with fuller capacity.

Annotator Bias. Knowing the correct answer causes annotators to rate the associated reasoning more generously—a well-documented outcome

bias. In our setting this is compounded by the scarcity of annotators who are both strong in Urdu and competent in advanced mathematics. κ identifies disagreement but cannot correct for shared blind spots.

5 Reflection

Our initial assumptions about this project were that to achieve better math reasoning in Urdu we needed math specific training information at the SFT level. The pipeline was formulated based on the following: CPT grounded language, SFT grounded in task format, and GRPO grounded in mathematical alignment. SFT was supposed to be scaffolding GRPO - not a reasoning contribution in its own right.

This view changed with the B3 result. A 1.5B model was only trained on general Urdu instruction data to achieve 31.2% EM on MATH-500, better than a model that was five times that size trained using chain-of-thought prompting. It also seems that the data of SFT training does not have an apparent effect on the structure of the responses it comprises of. This re-conceptualizes the low-resource bottleneck: the limited resource is not mathematical annotation but high-end structured instruction data in Urdu, which is more likely to be available by way of artful translation and community upkeep

This does not render domain-specific data to be irrelevant. The most difficult subcategories of MATH-500 are: olympean number theory, non-Euclidean geometry - this is where general education cannot be used to impart knowledge. B4 vs. B3 category will answer us where the structural alignment stops and domain knowledge is required. It is that frontier that is the most significant open question that we bring to the final assignment.

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